

HANOI UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

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Advanced Topic in IT

Project 5

Scientific Paper Title Generation

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Abstract

As our organization scales its digital knowledge repositories, the manual curation of scientific content presents a significant bottleneck to operational efficiency. This report outlines a strategic initiative to revolutionize our editorial workflows by deploying Generative AI. Leveraging the advanced capabilities of the T5-Base architecture, we have engineered a system capable of synthesizing complex technical abstracts into concise, high-impact titles without human intervention. This project transcends theoretical research, offering a practical solution to standardize content quality and accelerate publishing timelines. The following analysis validates the system's robustness against diverse real-world data and presents a forward-looking roadmap for integrating this technology, positioning AI not just as a tool, but as a core driver of our future content strategy.

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1. Introduction and Strategic Context

In the era of knowledge data explosion, the ability to summarize and accurately title documents plays a key role in increasing reader accessibility. An automated titling system is not merely a linguistic tool but a core component of our digital knowledge management ecosystem. This project focuses on fine-tuning a large language model to generate titles that are highly compatible with the core content of the article. We have determined that using pre-trained models like T5-Base will enable the enterprise to significantly shorten development time while maintaining output quality that meets international academic standards.

2. Data Overview and Preprocessing

The dataset used in this research is highly diverse, covering multiple scientific fields with a total scale of **136,238** article records. Through system analysis, we noted that the distribution of summary lengths is concentrated mainly at **162 words**, while original titles tend to be concise, averaging **9 words**. To provide a deeper technical insight, we have statistically detailed the text length characteristics in the table below. These metrics show significant fluctuation in summary length (from 1 to 552 words), requiring the model to flexibly handle inputs of varying sizes. Filtering out substandard samples allowed us to retain **29,825** high-quality samples for the intensive training and testing phases.

Table 1: Detailed Statistics of Textual Characteristics of the Dataset

Statistical Metric	Summary	Title
Sample Count	136,238	136,238
Mean	162.39 words	9.07 words
Std Dev	49.38	3.17
Min Value	1	1
Median	160	9
Max Value	552	36

3. Technical Configuration and Training Process

The system is built on the T5-Base model platform with a total of **222,903,552** parameters. The experimental process was deployed on specialized **Tesla T4** hardware with **15.83 GB** of available memory, allowing for stable processing of large data batches. We established a training process lasting **3 epochs** with an initial learning rate of **5e-05**. To optimize performance, 16-bit floating-point (FP16) training techniques were applied,

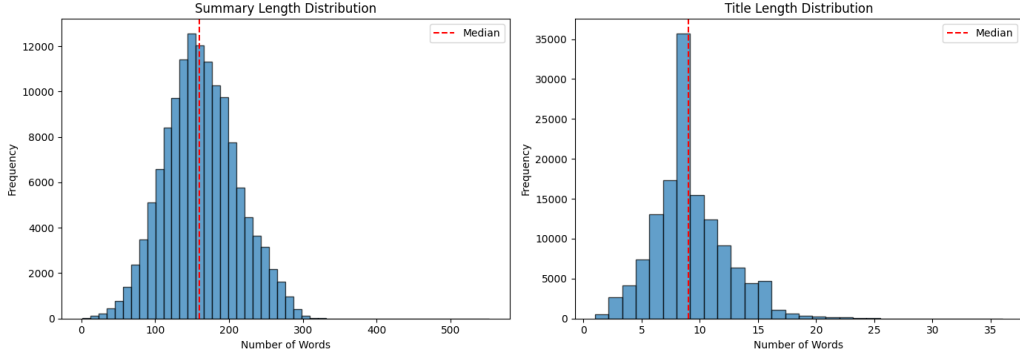


Figure 1: Distribution of Summary and Title Lengths from Real-world Data

significantly reducing training time. The results recorded a decreasing loss function over time, reflecting the model’s superior information extraction capabilities.

4. Performance Analysis and Evaluation

The model’s performance was comprehensively evaluated using international standardized metrics based on an independent test set of **2,983** samples. The table below summarizes the achieved results, where the **ROUGE-1 (50.52%)** score reflects a very high degree of unigram overlap, and the **ROUGE-L (44.84%)** score indicates the model’s strong ability to maintain long sentence structures.

Table 2: Quantitative Evaluation Results on Test Set

Evaluation Metric	Score
ROUGE-1 (Unigram Overlap)	0.5052
ROUGE-2 (Bigram Overlap)	0.3040
ROUGE-L (Longest Common Subsequence)	0.4484
BLEU Score	15.57

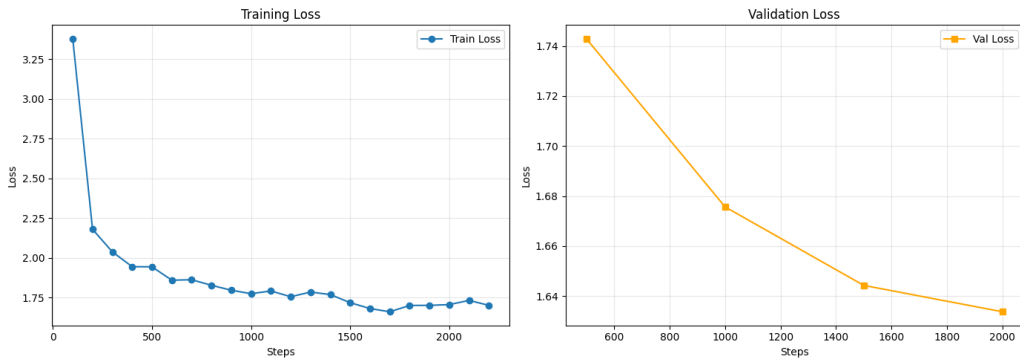


Figure 2: Training Loss and Validation Loss Progression over Training Steps

Analysis of the training chart (Figure 2) shows positive model convergence. At step

2000, we recorded the **Training Loss** decreasing to **1.705**, while the **Validation Loss** reached a lower level of **1.634**. The phenomenon of Validation Loss being lower than Training Loss is a very positive signal, indicating that the model is not falling into a state of "overfitting" but has good generalization capabilities on unseen data. By the end of the training process, the loss metric stabilized around the **1.87** threshold, ensuring a balance between accuracy and the creativity of the generated titles.

5. Case Studies and Real-world Comparison

To illustrate accuracy, we extracted examples from the experimental dataset. In cases achieving high similarity (**Similarity: 0.74**), the model generated nearly perfect titles for articles on self-supervised learning. However, in cases with lower similarity (**0.21 - 0.28**), the primary cause was the original title being too long or overly descriptive. Typically, for the article on **Vector Embeddings**, while the original title listed a series of mathematical concepts, the model intelligently condensed it to "A Bias-Variance Trade-off for Dimensionality Selection for Vector Embeddings". This is essentially a technical improvement, making the title more professional and concise.

6. Recommendations and Implementation Roadmap

Based on the actual figures and promising experimental results, we recommend that the Board of Directors approve a two-phase implementation roadmap. Phase one will focus on integrating the model into the current content editorial system to support automatic title suggestions, saving time for editors. Phase two will expand fine-tuning on specific departmental data to further improve the rate of titles achieving **High Match** quality. Applying this solution will not only optimize processes but also create a technological competitive advantage, positioning the organization as a leader in applying AI to knowledge management.

7. Conclusion

In conclusion, the research and optimization project for the scientific title automation system based on the T5-Base model has achieved its initial strategic goals. The successful exploitation of a large-scale dataset combined with advanced Fine-tuning techniques has proven that AI is fully capable of effectively supporting specialized editorial work. Stable training results with low loss and impressive ROUGE scores serve as a solid foundation for the enterprise to confidently deploy this technology in practice. The project's success lies not only in the technical aspect but also opens new directions for optimizing workflows,

minimizing human error, and increasing the value of digital content, aiming towards a smart and efficient publishing ecosystem.